Explainable Diabetes Risk Screening

A leakage-safe comparison of three machine-learning models on 253,680 CDC health-survey answers, with SHAP explanations and a working web app

Abdullah Mahmood

Student ID: i26043871

Many people have diabetes and don't know it

Blood test

needed for diagnosis (HbA1c / glucose) → requires a clinic visit

21 questions

a cheap survey anyone can answer in two minutes

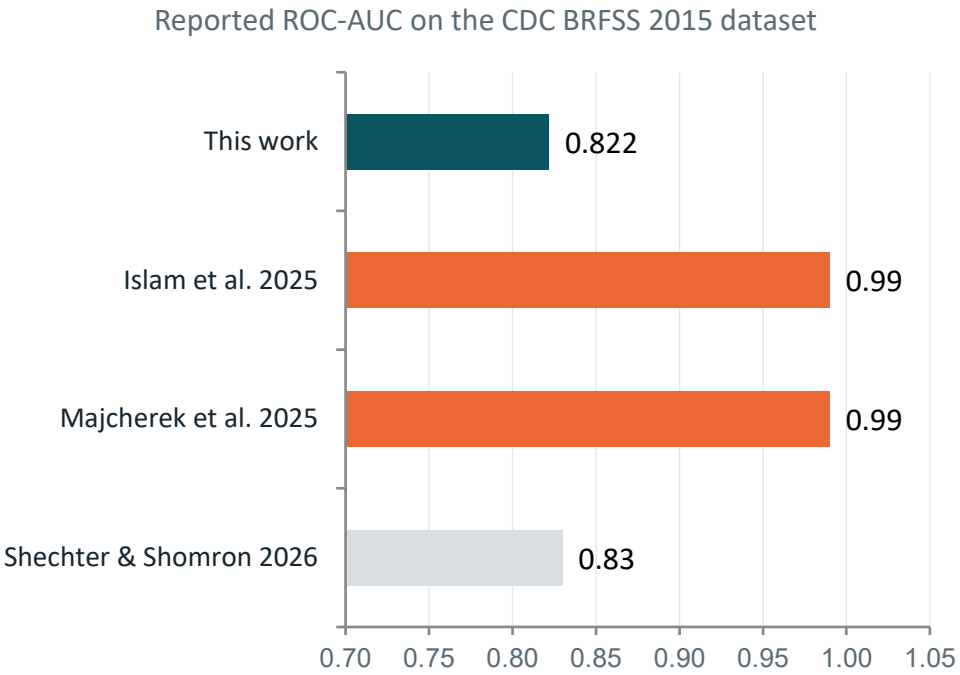
Can ML help?

rank people by risk, refer the high-risk ones for a test

Why it matters

- Undiagnosed diabetes leads to heart, kidney and eye damage that early action can delay
- Population surveys (CDC BRFSS) already collect hundreds of thousands of answers each year
- A screening tool must be honest about its accuracy and explain why it flags someone

Same dataset, very different published results



The gap

- 0.99 results used oversampling (SMOTE); when it happens vs. the split is often not reported
- Duplicates and error analysis rarely discussed

My objectives

- Honest, leakage-safe comparison of 3 models
- Explain predictions with SHAP + analyse errors
- Working web prototype

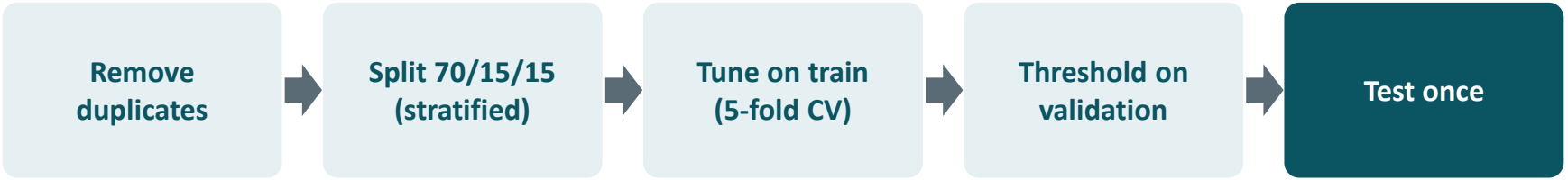
Dataset and leakage-safe pipeline

253,680
survey answers (CDC BRFSS 2015, UCI #891)

21
health & lifestyle features

13.9%
have diabetes (imbalanced)

9.5%
exact duplicates → removed



Every step that learns from data uses training data only. Two new features: BMI category (WHO) and number of chronic conditions.

Three models, from simple to powerful

1

Logistic Regression

Baseline

A weighted sum of the answers. Fully transparent: each weight shows the direction of risk.

2

Random Forest

Bagging

Hundreds of decision trees vote. Captures non-linear patterns.

3

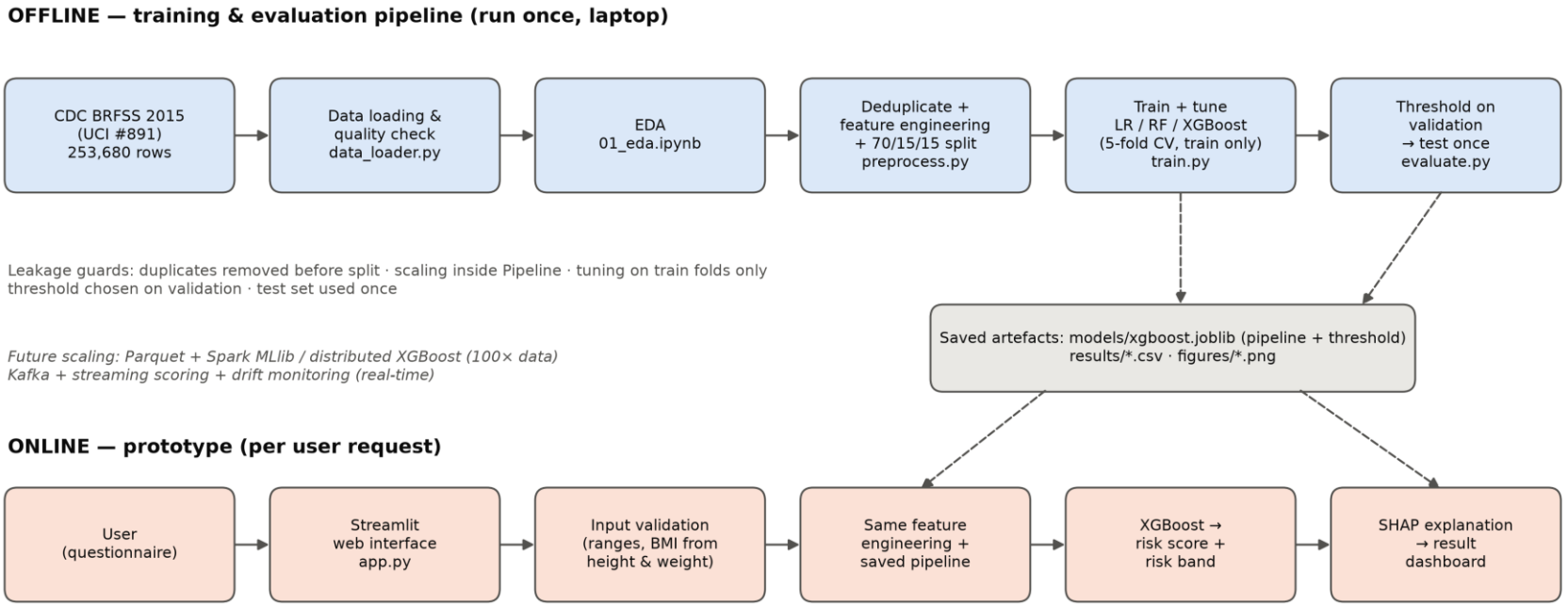
XGBoost

Boosting

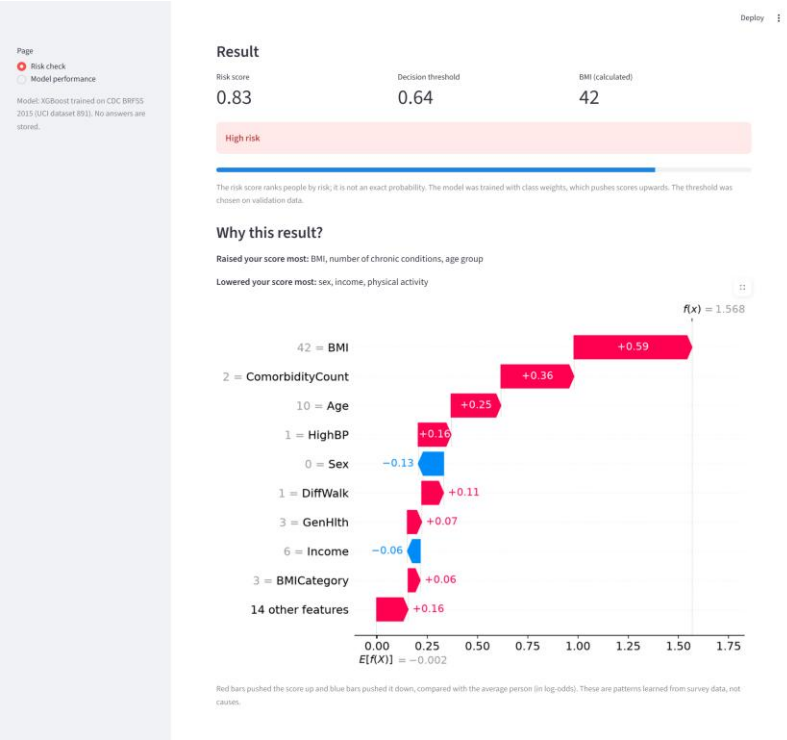
Trees built one after another, each fixing the last one's mistakes. Usually best on tables.

Imbalance: class weights (mistakes on diabetics count ~5.5× more) — no synthetic rows, so nothing can leak. No deep learning: on 21 tabular features, trees match it and are easier to explain.

System architecture



Live demo: Streamlit web app

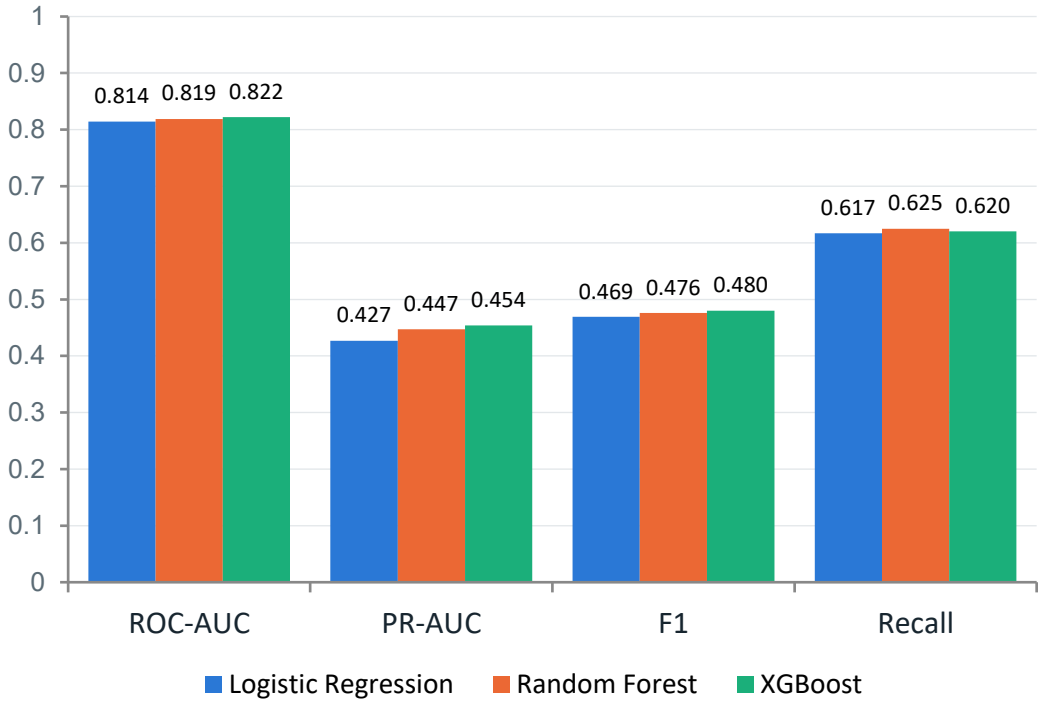


Demo script

1. Default person → score 0.37 → Low risk
2. Change: age 65-69, weight 120 kg, high BP, high cholesterol, difficulty walking → score 0.83 → High risk
3. Read "Why this result?": BMI, chronic conditions, age raised the score
4. Show the Model performance page

Run: streamlit run app/app.py · Backup: screenshots in figures/

Results on the untouched test set (34,422 people)



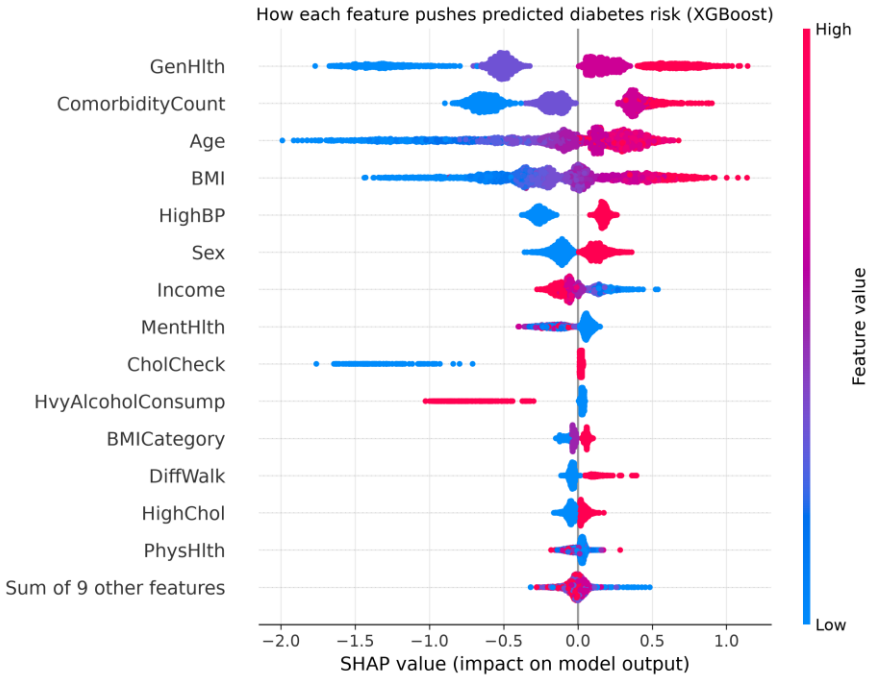
0.822

ROC-AUC (XGBoost) — matches leakage-safe study (0.83)

84.7%

accuracy of a model that says "no" to everyone — finds 0 diabetics

What drives the model, and who it misses



SHAP: top drivers

General health, number of chronic conditions, age, BMI, high blood pressure — match medical knowledge

Error analysis

84%

of misses are also missed by the other 2 models

0.28

recall for ages 18–44 (missed most)

Missed people look healthy on paper: good self-rated health, normal BMI, no high BP. The limit is the data (no blood tests), not the algorithm.

Conclusion

1. **Honest** — ROC-AUC 0.822 with a leakage-safe protocol — matches comparable published work
2. **Explainable** — SHAP for every prediction; 10 features keep almost all accuracy
3. **Working** — Streamlit app with risk bands, explanations and a clear disclaimer

Limitations: self-reported 2015 US data, no lab values, lower recall for young adults, scores not calibrated.
Future work: other BRFSS years, calibration, prediabetes as a 3rd class, a 12-question version.